

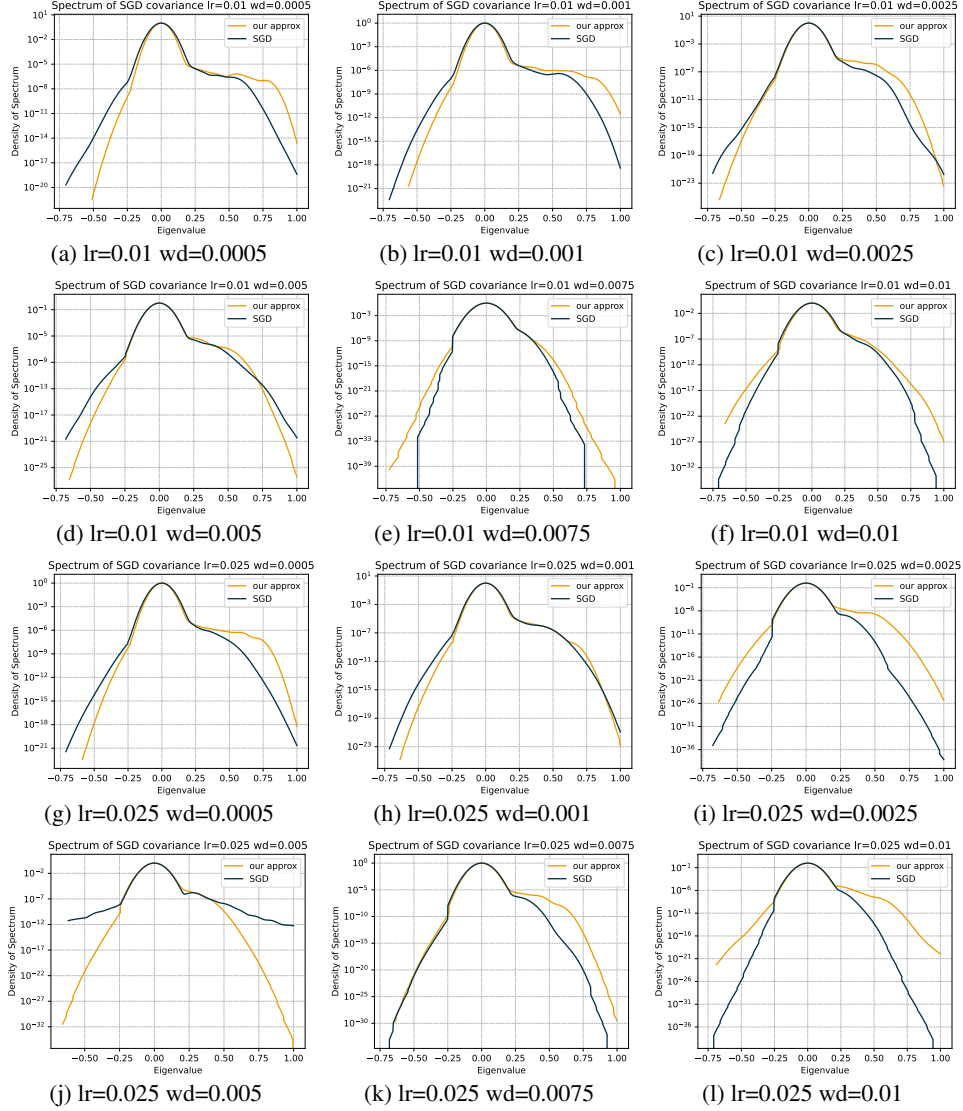


Figure 17: We use Stochastic Lanczos Quadrature to estimate the empirical spectral density of the SGD covariance, with and without the decoupling approximation, during the large-LR phase. Experiments with ResNet-18 on the CIFAR-10 dataset, varying learning rate and weight decay, show a substantial overlap in the two spectra, validating our approximation.

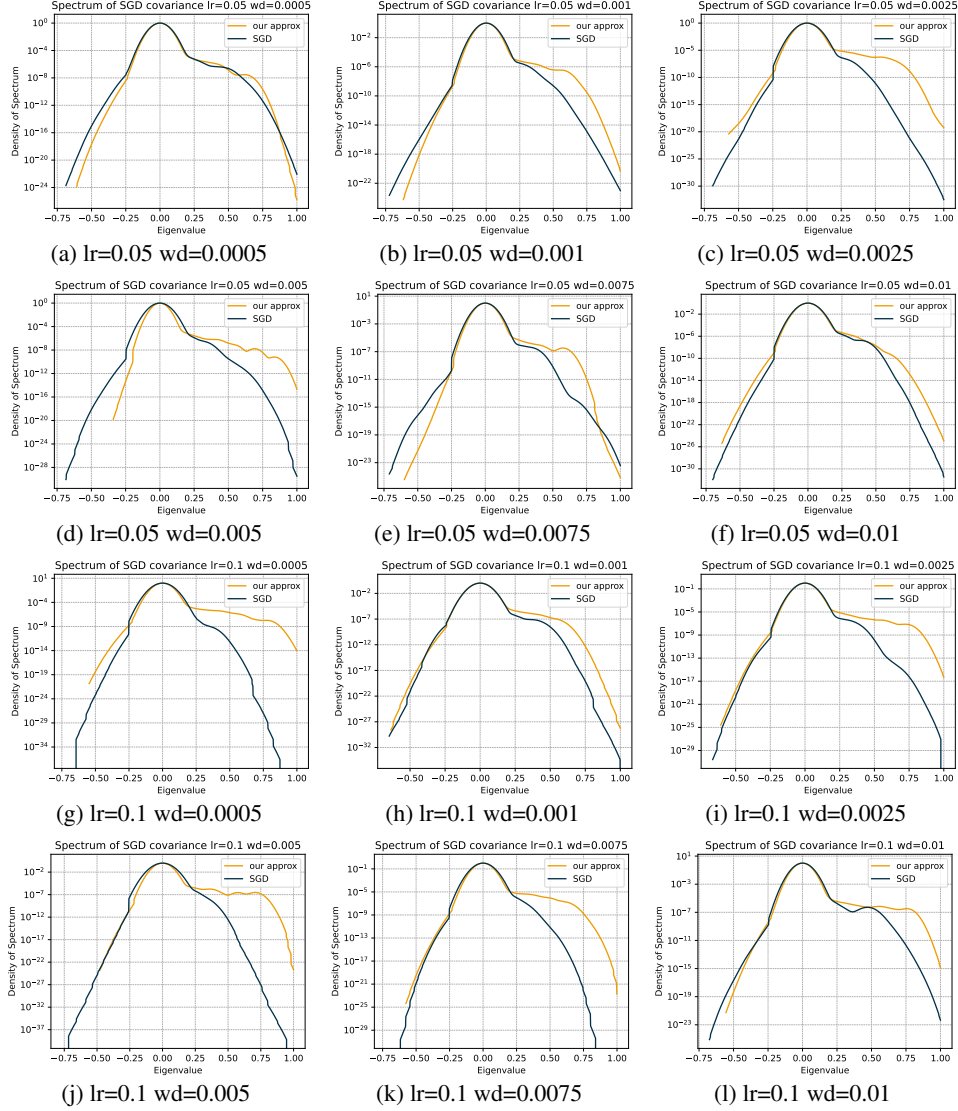


Figure 18: We use Stochastic Lanczos Quadrature to estimate the empirical spectral density of the SGD covariance, with and without the decoupling approximation, during the large-LR phase. Experiments with ResNet-18 on the CIFAR-10 dataset, varying learning rate and weight decay, show a substantial overlap in the two spectra, validating our approximation.

C.7 Empirical verification of the conjecture through snapshot ensembles

To further validate our conjecture, we performed additional experiments. Specifically, within the same ResNet18 on CIFAR-10 setting as in our main experiments, we created snapshot ensembles (Huang et al., 2017) by averaging in function space along the SGD trajectory every 10 epochs for the combinations of learning rate (LR) and weight decay (WD) considered in the paper.

To assess whether the mean of the stationary distribution in function space aligns closely with the EMA, where the Jacobian norm is regularized, we compared the performance of snapshot ensembles with that of the EMA. Additionally, we computed the Total Variation Distance $\mathcal{D}_{TV}$ between the softmax outputs of the ensemble and the EMA on the Test set

$$\mathcal{D}_{TV} = \frac{1}{2N} \sum_{i=1}^N \sum_{j=1}^C \left| p_{\text{ensemble},j}^{(i)} - p_{\text{EMA},j}^{(i)} \right|.$$

The results in Table 2 show a strong alignment in test accuracies, while those in Table 3 indicate a low Total Variation across all combinations. Together, these findings offer further validation for our conjecture.

Table 2: Test Error for Snapshot Ensemble and EMA for different values of learning rate (LR) and weight decay (WD).

WD	LR=0.001		LR=0.005		LR=0.01		LR=0.025		LR=0.05		LR=0.1		LR=0.15	
	ENS	EMA	ENS	EMA	ENS	EMA	ENS	EMA	ENS	EMA	ENS	EMA	ENS	EMA
0.0000	0.32	0.33	0.27	0.26	0.24	0.25	0.17	0.17	0.17	0.17	0.13	0.13	0.13	0.13
0.0005	0.32	0.32	0.29	0.29	0.24	0.24	0.18	0.17	0.13	0.16	0.13	0.13	0.11	0.13
0.0010	0.32	0.33	0.25	0.27	0.21	0.21	0.13	0.19	0.10	0.13	0.10	0.11	0.11	0.11
0.0015	0.32	0.34	0.23	0.22	0.22	0.25	0.11	0.14	0.10	0.12	0.10	0.10	0.09	0.09
0.0025	0.30	0.30	0.22	0.22	0.19	0.20	0.09	0.10	0.10	0.11	0.10	0.10	0.10	0.11
0.0050	0.33	0.34	0.21	0.20	0.12	0.16	0.10	0.10	0.10	0.10	0.09	0.09	0.10	0.09
0.0075	0.35	0.37	0.15	0.16	0.10	0.11	0.11	0.11	0.09	0.08	0.10	0.09	0.12	0.10
0.0100	0.31	0.34	0.13	0.15	0.11	0.11	0.10	0.10	0.11	0.10	0.11	0.09	0.13	0.13

Table 3: Total Variation Distance between softmax output of Ensemble and EMA.

WD	LR=0.001	LR=0.005	LR=0.01	LR=0.025	LR=0.05	LR=0.1	LR=0.15
0.0000	0.03	0.02	0.01	0.01	0.01	0.01	0.01
0.0005	0.04	0.02	0.02	0.03	0.09	0.08	0.07
0.0010	0.04	0.05	0.04	0.10	0.09	0.07	0.07
0.0015	0.04	0.07	0.07	0.08	0.08	0.07	0.07
0.0025	0.04	0.11	0.10	0.06	0.08	0.08	0.09
0.0050	0.06	0.15	0.11	0.09	0.08	0.10	0.11
0.0075	0.08	0.12	0.10	0.10	0.10	0.11	0.12
0.0100	0.10	0.09	0.10	0.10	0.11	0.12	0.13